

D799 - Task 1 - Attempt 1

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Mobile Application Creation Plan

A1. Problems the Application Will Address

Students often struggle with fragmented access to academic resources, requiring them to check multiple platforms for schedules, grades, announcements, and peer communication. This results in missed deadlines, reduced engagement, and inefficient time management (Kumar Basak, Wotto, & Bélanger, 2018). The proposed mobile application, Campus Connect, addresses this problem by consolidating these resources into a single platform, enabling students to manage their academic and social campus life more effectively.

A2. Primary Features of the Application

- Class Schedule Viewer: Displays daily and weekly class schedules.
- Grades and Announcements: Provides quick access to updated grades and important course announcements.
- Messaging Forum: Allows communication with peers and professors in a simplified forum or chat.
- Notifications: Sends alerts for upcoming classes, grade postings, and announcements.

A3. User Growth

The app will initially target university students and professors within a single campus environment. As adoption increases, scalability will allow integration across multiple institutions

and expanded features such as event calendars, library resources, and student clubs. The architecture will support increased user traffic by allowing modular expansion of backend services without requiring redesign of the core system.

A4. Data Management

Data will be stored locally on devices using SQLite for offline availability, with synchronization to a secure cloud database for backup and cross-device access. User accounts, schedules, and messaging data will be encrypted during transmission and storage. API endpoints will be designed to ensure efficient retrieval of only the most relevant and updated information, minimizing unnecessary data flow.

A5. Performance Considerations

The application will be optimized to maintain fast load times and smooth navigation, even on lower-powered devices. Caching will be used to minimize repeated data requests, and asynchronous data loading will ensure that users can continue navigating the app while background updates occur. Load testing will validate that the app can handle expected peak usage, such as the start of a semester when schedules and announcements are frequently accessed.

A6. Resource Optimization

- Minimize battery usage by limiting background processes and push notifications to essential updates.
- Optimize storage by compressing data and retaining only essential offline records.

- Use efficient APIs to reduce mobile data consumption for students with limited connectivity.

A7. Backend Architecture

The backend will be built on a .NET MAUI cross-platform framework, ensuring compatibility with both Android and iOS devices

- SQLite for local storage.
- Cloud-based API services for synchronization and data sharing.
- Authentication and access control to secure user accounts.
- Modular design enabling future integration of additional features such as event tracking, library access, or academic advising.

B1. User Persona 1 – Undergrad Student

Demographics

- Age: 20 years old
- Role: Full-time sophomore student
- Technical Experience Level: Intermediate; familiar with mobile apps and online portals
- Daily Routines: Attends multiple classes daily, checks email/portals for updates, studies in the library, and participates in evening clubs.

Needs, Goals, and Challenges

- Need: Centralized platform to view schedules, grades, and announcements.

- Goal: Stay on top of assignments and deadlines to improve academic performance.
- Challenge: Overwhelmed by scattered information across multiple systems, often missing updates.

How the Application Meets Them

- Schedule Viewer organizes daily/weekly classes in one place.
- Grades & Announcements consolidates updates with push notifications.
- Messaging Forum reduces reliance on separate group chats, ensuring critical communication isn't missed.

B2. User Persona 2 - Professor

Demographics

- Age: 45 years old
- Role: University professor teaching multiple courses
- Technical Experience Level: Advanced; proficient with digital grading tools, email, and LMS platforms
- Daily Routines: Prepares and delivers lectures, manages grading, answers student inquiries, and posts announcements.

Needs, Goals, and Challenges

- Need: Reliable way to deliver announcements and reminders directly to students.

- Goal: Improve class communication and reduce time spent clarifying missed information.
- Challenge: Students frequently overlook email communications, causing confusion and wasted effort.

How the Application Meets Them

- Announcements Feature ensures updates are delivered directly to students' mobile devices.
- Push Notifications make messages timely and harder to miss.
- Messaging Forum centralizes Q&A, cutting down repetitive emails.

C1. Sensors and Communication

The application will depend primarily on Wi-Fi and cellular networks to allow students and professors to access schedules, grades, announcements, and messages in real time. Mobile device notification services will be used to push reminders about upcoming classes, deadlines, and posted grades. Calendar integration can also allow the app to sync course schedules directly into the device's native calendar. As an optional feature, Bluetooth communication may be used for local peer-to-peer sharing of notes or files when users are in close proximity.

C2. Technical Limitations

The app's performance is partially dependent on stable internet connectivity. Students in areas with limited or inconsistent service may face delays in receiving updates or syncing information across devices (Kumar Basak, Wotto, & Bélanger, 2018). Offline functionality will allow users to view already downloaded schedules and grades, but new announcements or changes will not be

reflected until connectivity is restored. Device-level limitations such as low memory, slower processors, or restricted storage space could also affect performance if the app is not optimized to cache and compress data efficiently.

C3. Security and Privacy

Because the application will manage sensitive academic information, security and privacy are critical design considerations. All communication between the app and backend services will be encrypted to prevent data interception. User authentication will be required through secure login credentials tied to existing student accounts. The app will minimize data collection to only what is essential, such as schedules, announcements, and grades, and will require user consent before any data is shared outside of the platform. Access controls will also be implemented to ensure that professors can post announcements while students are limited to viewing or participating in discussions, reducing the risk of unauthorized changes.

C4. Seamless Connectivity

The application will be designed to maintain a smooth user experience across different connectivity environments. Offline caching will ensure that students can continue to access key information such as schedules and recently viewed grades without an active connection. Once connectivity is restored, the app will automatically synchronize updates in the background without requiring manual refreshes from the user.

D1. Data Collection

The application will limit data collection to only what is necessary to provide its core functions, including class schedules, grades, announcements, and communication records between students

and professors. No unnecessary personal data such as location history, contacts, or unrelated app activity will be gathered. By restricting collection, the app reduces ethical risks and respects the privacy of its users while still enabling essential academic support.

D2. Data Storage

All collected information will be stored securely using a combination of local and cloud-based solutions. Local storage will be managed through an encrypted SQLite database on the device to support offline access, while cloud storage will allow synchronization across multiple devices and backup in case of device failure. Both storage systems will follow secure coding practices to prevent unauthorized access, ensuring that sensitive information like grades and communications remain protected.

D3. Data Sharing

Data sharing will be limited to the minimum necessary for functionality. Professors will be able to post announcements and grades that are shared only with their students, and students may participate in peer discussions through controlled forums. No third-party organizations will have access to user data without explicit permission. Only anonymized or role-specific data will be shared through secure APIs to avoid exposing personal information.

D4. User Consent

User consent will be obtained before any data is collected, stored, or shared. During account creation, users will be presented with clear terms of service and privacy agreements outlining what data is gathered and how it will be used. Consent will not be hidden in technical jargon but written in accessible language to ensure students and faculty can make informed decisions.

Additionally, users will have the option to withdraw consent for non-essential features, such as peer-to-peer data sharing, without losing access to the core functionality of the application.

E1. Processing Limitations

The application is designed to operate efficiently on a wide range of mobile devices, including those with lower processing capabilities. Since students often use mid-range or budget devices, the app will minimize background processing and avoid unnecessary tasks that could slow performance. Complex operations, such as generating reports or syncing large amounts of data, will be handled by the server rather than relying heavily on device resources, reducing strain on user hardware.

E2. Data Flow and Traffic

Data flow within the application will follow a lightweight and optimized pattern to prevent excessive network usage. User actions, such as checking schedules or grades, will trigger targeted requests to the server, while frequently accessed information will be cached locally to reduce repeated traffic. Push notifications will be used instead of constant background polling, ensuring that updates are delivered efficiently without placing unnecessary demand on the network.

E3. Data Handling Strategy

The app will adopt a hybrid data handling strategy, combining local storage for offline access with cloud synchronization for multi-device use. Data will be structured in a way that ensures

quick retrieval and updates, with timestamps used to identify the most current version of records. Error handling will be implemented to manage incomplete transfers or network interruptions, ensuring that no duplicate or inconsistent data is presented to users.

E4. Processing Power

While most day to day functions will require minimal processing power, the system will be able to scale when handling larger volumes of users or data. The backend architecture will take advantage of server-side resources for tasks such as batch updates, analytics, or data validation, ensuring that mobile devices remain responsive and energy-efficient. This balance allows the app to perform smoothly on personal devices while still supporting institutional growth and increased user demand over time.

F1. User Interface (UI) Design

The interface of the application will be simple, clean, and visually consistent so that users can navigate it without confusion. The home screen will serve as a central hub that provides quick access to schedules, grades, announcements, and messaging. Fonts, colors, and button sizes will be chosen to maximize readability and accessibility for both students and professors (Dabbagh & Kitsantas, 2012). The design will avoid clutter, focusing only on essential features so that even users with minimal technical skills can interact with the app effectively.

F2. Information Architecture

The information architecture will be organized in a logical and hierarchical manner to help users find information quickly. The home dashboard will provide direct links to core functions such as viewing a class schedule, checking grades, and accessing announcements. From each main

section, subpages will allow for detailed exploration, such as viewing course-specific grades or messages related to a single class. This layered but consistent structure ensures that information is not buried and that users always know where they are within the app.

F3. Usability Testing

To validate the design, usability testing will be conducted with both students and professors before deployment. Participants will be asked to complete common tasks such as checking their schedule, finding a grade update, or posting an announcement. Their feedback will be collected to identify confusing elements or navigation issues. Iterative adjustments will be made based on test results to ensure the app is intuitive, responsive, and efficient. This process will help confirm that the design supports real-world academic workflows and meets the needs of its intended audience.

G. Device Interactivity

The application can benefit from device interactivity by connecting with other systems and devices that students and professors already use. One example is calendar integration, where the app can sync course schedules and deadlines directly into the mobile device's native calendar, ensuring users receive reminders even outside the app. Push notification services on the device will also be leveraged to alert users about class changes, posted grades, or new announcements in real time (Dabbagh & Kitsantas, 2012). In addition, the app can interact with cloud-based learning platforms through secure APIs, allowing grades and course updates to flow seamlessly between institutional systems and the mobile application. Limited use of Bluetooth can enable local peer-to-peer interactions, such as sharing notes or resources between students without

relying on an internet connection. These features enhance usability by reducing redundancy and making academic information more accessible across devices.

H. Scenario of Device Interaction

A common scenario where the application interacts with another device is when a student adds a new course or assignment deadline within the app and it automatically syncs to their phone's native calendar. Once synced, the mobile device's calendar generates reminders and alarms based on the user's settings, ensuring the student does not miss important events. For example, if a professor posts an announcement about an upcoming exam, the app pushes that information directly to the student's device, and the device in turn delivers a timed notification even if the app is closed. This seamless interaction between the mobile application and the device's calendar and notification system helps keep students organized without requiring additional manual input.

I. Application Value

The application provides value to users by centralizing academic resources into a single, user-friendly platform that reduces the need to navigate multiple portals and systems. For students, the app saves time, improves organization, and ensures they never miss critical deadlines or updates (Kumar Basak, Wotto, & Bélanger, 2018). For professors, it simplifies communication with students and reduces the number of repeated questions and missed announcements. Users would be willing to pay for the application because it directly supports academic success by improving time management and reducing stress. The convenience of having schedules, grades, announcements, and messaging all in one place justifies the cost, especially when compared to the inefficiencies and frustrations of relying on multiple disjointed tools (Dabbagh & Kitsantas, 2012).

J. Communication Protocols

The application will primarily rely on Wi-Fi and cellular data connections to ensure continuous access to schedules, grades, and announcements. The advantage of Wi-Fi and mobile networks is that they provide broad coverage and allow real-time synchronization across devices, while the limitation is that performance may decrease in areas with poor connectivity or limited data plans. Security measures for these connections will include HTTPS with TLS encryption to protect sensitive academic data during transmission.

Bluetooth can be implemented for optional peer-to-peer interactions such as sharing notes or small files between nearby students. The advantage of Bluetooth is that it does not require internet access, making it useful in classrooms or study groups, but it is limited by short range and slower data transfer speeds compared to Wi-Fi. To secure Bluetooth communication, the app will implement device pairing and encrypted channels to prevent unauthorized access.

Near-field communication (NFC) could also be supported in future iterations of the app for quick device-to-device interactions, such as exchanging contact or course group details with a simple tap. The advantage of NFC is its ease of use and very low energy consumption, but it is limited by extremely short range and slower speeds. Security measures for NFC would include encryption and user confirmation before any data exchange occurs.

K. Mitigation Measures

The application will be built on a cross-platform framework so that both Android and iOS devices can run it reliably. Regular updates and testing across different operating system versions and device models will help reduce the risk of incompatibility. Maintaining stable connectivity

will be achieved through lightweight data requests, efficient APIs, and the use of push notifications instead of constant background polling, which reduces the likelihood of connection drops while minimizing battery and data usage. Data synchronization challenges will be addressed through the use of timestamps and conflict resolution rules to ensure that the most recent information is always displayed, even when multiple devices are accessing the same account. In cases of interrupted connections, offline caching will allow the user to continue working with stored data until the app can reconnect and synchronize changes in the background.

References

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